```
!pip -q install pypdf sentence-transformers google-genai
```

```
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 402.7/402.7 kB 1.8 MB/s eta 0:00:00
```

```
from google.colab import userdata
from google import genai

client = genai.Client(
    api_key=userdata.get("GEMINI_APIKEY")
)

print("Gemini API client is ready.")
```

```
Gemini API client is ready.
```

```
from google.colab import files
from pypdf import PdfReader

uploaded = files.upload()
pdf_path = next(iter(uploaded))

reader = PdfReader(pdf_path)
text = "\n".join(page.extract_text() or "" for page in reader.pages)

print("Total pages:", len(reader.pages))
print("Total extracted characters:", len(text))
print("\nFirst 500 characters:\n", text[:500])
```

```
Choose Files sample_stu…s_policy.pdf
sample_student_services_policy.pdf(application/pdf) - 32085 bytes, last modified: 10/13/2026 - 100% done
Saving sample_student_services_policy.pdf to sample_student_services_policy.pdf
Total pages: 5
Total extracted characters: 4811

First 500 characters:
 Student Services and Campus Facilities Policy
Fictional sample document for educational use
Sample policy document for a fictional institution. Institution: Greenfield Institute of Applied Learning.
Policy version: 2.1. Effective date: 1 July 2026. Review date: 30 June 2027.
This document describes selected student-service procedures and campus rules. It is a fictional training
document created for a retrieval-augmented generation (RAG) assignment. The facts in this document
apply only to this s
```

```
def make_chunks(text, chunk_size=800, overlap=150):
    chunks = []
    start = 0

    while start < len(text):
        end = start + chunk_size
        chunk = text[start:end].strip()

        if chunk:
            chunks.append(chunk)

        start += chunk_size - overlap

    return chunks

chunks = make_chunks(text, chunk_size=800, overlap=150)

print("Chunk size: 800")
print("Overlap: 150")
print("Number of chunks:", len(chunks))
print("\nFULL EXAMPLE CHUNK:\n")
print(chunks[0])
```

```
Chunk size: 800
Overlap: 150
Number of chunks: 8

FULL EXAMPLE CHUNK:
```

```
Student Services and Campus Facilities Policy
Fictional sample document for educational use
Sample policy document for a fictional institution. Institution: Greenfield Institute of Applied Learning.
Policy version: 2.1. Effective date: 1 July 2026. Review date: 30 June 2027.
This document describes selected student-service procedures and campus rules. It is a fictional training
document created for a retrieval-augmented generation (RAG) assignment. The facts in this document
apply only to this sample institution.
The Student Services Office is located in Block B, Room 104. Office hours are Monday to Friday, 9:00
a.m. to 4:30 p.m., excluding declared campus holidays. Students must show their campus ID card when
requesting a service in person.
The official email for student service requests
```

```
from sentence_transformers import SentenceTransformer
import numpy as np

embedding_model = SentenceTransformer(
    "sentence-transformers/all-MiniLM-L6-v2"
)

chunk_embeddings = embedding_model.encode(
    chunks,
    convert_to_numpy=True,
    normalize_embeddings=True
)

print("Number of chunk texts:", len(chunks))
print("Embedding matrix shape:", chunk_embeddings.shape)
print("Embeddings and their corresponding chunk texts are ready.")
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fast
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN t
modules.json: 100%  349/349 [00:00<00:00, 21.4kB/s]
config_sentence_transformers.json: 100%  116/116 [00:00<00:00, 5.79kB/s]
README.md: 100%  10.5k/10.5k [00:00<00:00, 496kB/s]
sentence_bert_config.json: 100%  53.0/53.0 [00:00<00:00, 4.88kB/s]
config.json: 100%  612/612 [00:00<00:00, 47.2kB/s]
model.safetensors: reconstructing file: 100%  90.9MB / 90.9MB, 8.48MB/s
model.safetensors: downloading bytes:  85.0MB, 7.59MB/s
Loading weights: 100%  103/103 [00:00<00:00, 2490.64it/s]
tokenizer_config.json: 100%  350/350 [00:00<00:00, 33.6kB/s]
vocab.txt: 100%  232k/232k [00:00<00:00, 7.47MB/s]
tokenizer.json: 100%  466k/466k [00:00<00:00, 19.2MB/s]
special_tokens_map.json: 100%  112/112 [00:00<00:00, 8.34kB/s]
config.json: 100%  190/190 [00:00<00:00, 13.4kB/s]
Number of chunk texts: 8
Embedding matrix shape: (8, 384)
Embeddings and their corresponding chunk texts are ready.
```

```
def retrieve(question, chunks, embeddings, top_k=3):
    question_embedding = embedding_model.encode(
        [question],
        convert_to_numpy=True,
        normalize_embeddings=True
    )[0]

    scores = embeddings @ question_embedding
    indices = np.argsort(scores)[::-1][:top_k]

    return [(int(i), float(scores[i]), chunks[i]) for i in indices]

question = "How much does a replacement campus ID card cost?"

results = retrieve(question, chunks, chunk_embeddings)

for rank, (idx, score, chunk) in enumerate(results, start=1):
    print(f"\nRank {rank} | Chunk {idx} | Similarity: {score:.4f}")
    print(chunk)
    print("-" * 60)
```

```
Rank 1 | Chunk 0 | Similarity: 0.4829
Student Services and Campus Facilities Policy
Fictional sample document for educational use
Sample policy document for a fictional institution. Institution: Greenfield Institute of Applied Learning.
Policy version: 2.1. Effective date: 1 July 2026. Review date: 30 June 2027.
This document describes selected student-service procedures and campus rules. It is a fictional training
document created for a retrieval-augmented generation (RAG) assignment. The facts in this document
apply only to this sample institution.
The Student Services Office is located in Block B, Room 104. Office hours are Monday to Friday, 9:00
a.m. to 4:30 p.m., excluding declared campus holidays. Students must show their campus ID card when
requesting a service in person.
The official email for student service requests
------------------------------------------------------------

Rank 2 | Chunk 5 | Similarity: 0.4352
umbing, electrical, or furniture problems to
the hostel office. Urgent electrical faults must also be reported immediately to the duty warden. Students
must not attempt electrical repairs themselves.
The hostel maintenance desk operates Monday to Saturday from 9:30 a.m. to 4:00 p.m. Routine
maintenance requests are normally reviewed within 3 working days. Review time is not a guarantee that
the repair will be completed within that period.

Fees, complaints, and emergency contacts
A replacement campus ID card costs ■150. Students must submit the replacement request form and
show another valid photo ID. The usual processing time is 3 working days.
A printed transcript request costs ■100 for the first copy and ■40 for each additional copy ordered in the
same request. Transcript requests are s
------------------------------------------------------------

Rank 3 | Chunk 6 | Similarity: 0.4238
ys.
A printed transcript request costs ■100 for the first copy and ■40 for each additional copy ordered in the
same request. Transcript requests are submitted to the Student Services Office. The published
processing time is 5 working days after payment confirmation.
Students may submit a formal complaint using the complaint form available at the Student Services
Office. The office acknowledges a formal complaint within 2 working days and aims to provide an initial
decision within 10 working days. If more investigation is required, the student receives an update
explaining the delay.
For a campus medical emergency, call the campus response desk at 044-5550-0199 or contact the
nearest campus security officer. This number belongs only to the fictional sample institution and is not a
real emer
------------------------------------------------------------
```

```
def answer_question(question, chunks, embeddings, top_k=3):
    results = retrieve(question, chunks, embeddings, top_k)

    context = "\n\n".join(
        f"[Chunk {idx}]\n{chunk}"
        for idx, score, chunk in results
    )

    prompt = f"""
You are a question-answering assistant for one PDF document.

Answer using ONLY the supplied context.
Do not use outside knowledge or make assumptions.
If the context does not explicitly contain enough information
to answer the question, reply exactly:
"I cannot find the answer in the provided document."

Cite the supporting chunk as [Chunk N] when answering.
If the answer is unavailable, do not invent a citation.

CONTEXT:
{context}

QUESTION:
{question}

Answer:
"""

    response = client.models.generate_content(
        model="gemini-3.8-flash",
        contents=prompt
    )

    print("QUESTION:", question)
    print("\nANSWER:\n", response.text)
    print("\nRETRIEVED CHUNKS AND SCORES:")

    for idx, score, chunk in results:
        print(f"\nChunk {idx} | Similarity: {score:.4f}")
        print(chunk[:250].replace("\n", " "), "...")
```

```
import time
from google.genai.errors import ServerError

def answer_question_with_retry(q, chunks, chunk_embeddings, retries=3, delay=2):
    for attempt in range(retries):
        try:
            answer_question(q, chunks, chunk_embeddings)
            return
        except ServerError as e:
            if e.code == 503 and attempt < retries - 1:
                print(f"\n[Model busy, retrying in {delay} seconds... (Attempt {attempt+1}/{retries})]")
                time.sleep(delay)
                delay *= 2
            else:
                raise e

in_document_questions = [
    "How much does a replacement campus ID card cost?",
    "How many books can an undergraduate borrow, and for how many days?",
    "How long does a printed transcript request take to process?"
]

out_document_questions = [
    "What is the semester tuition fee?",
    "How many sick days may a hostel resident take each year?",
    "What is the penalty for arriving late to a computer laboratory?"
]

print("=" * 70)
print("IN-DOCUMENT QUESTIONS")
print("=" * 70)

for q in in_document_questions:
    answer_question_with_retry(q, chunks, chunk_embeddings)
    print("\n" + "=" * 70)

print("\nOUT-OF-DOCUMENT QUESTIONS")
print("=" * 70)

for q in out_document_questions:
```

```
    answer_question_with_retry(q, chunks, chunk_embeddings)
    print("\n" + "=" * 70)
```

```
Student Services and Campus Facilities Policy Fictional sample document for educational use Sample policy document for a f

======================================================================

OUT-OF-DOCUMENT QUESTIONS
======================================================================
QUESTION: What is the semester tuition fee?

ANSWER:
 I cannot find the answer in the provided document.

RETRIEVED CHUNKS AND SCORES:

Chunk 1 | Similarity: 0.4131
eclared campus holidays. Students must show their campus ID card when requesting a service in person. The official email f

Chunk 0 | Similarity: 0.3747
Student Services and Campus Facilities Policy Fictional sample document for educational use Sample policy document for a f

Chunk 6 | Similarity: 0.3457
ys. A printed transcript request costs ■100 for the first copy and ■40 for each additional copy ordered in the same reques

======================================================================
QUESTION: How many sick days may a hostel resident take each year?

ANSWER:
 I cannot find the answer in the provided document.

RETRIEVED CHUNKS AND SCORES:

Chunk 4 | Similarity: 0.4551
rs access is limited to a maximum of 2 hours per booking. Students must leave the room by the approved end time.  Hostel a

Chunk 5 | Similarity: 0.4196
umbing, electrical, or furniture problems to the hostel office. Urgent electrical faults must also be reported immediately

Chunk 3 | Similarity: 0.3400
to Friday. Access outside these hours requires written approval from the laboratory coordinator at least 1 working day in

======================================================================
QUESTION: What is the penalty for arriving late to a computer laboratory?

ANSWER:
 I cannot find the answer in the provided document.

RETRIEVED CHUNKS AND SCORES:

Chunk 2 | Similarity: 0.6274
```